

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
(3) Answers to the **same question** must be **written together**.
(4) Numbers to the **right** indicate **marks**.
(5) Draw **neat labeled diagrams** wherever **necessary**.
(6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. What does the term "Big Data" mean? Enumerate the various applications of Big Data.
 - b. Provide a concise explanation in a short note regarding the CAP theorem.
 - c. Describe MongoDB and elaborate on its features.
 - d. Provide a brief explanation of the key distinctions between MongoDB and SQL.

2. **Attempt any two of the following:** 12
 - a. Provide a detailed analysis of Binary JSON (BSON), covering its structure, use cases, and advantages.
 - b. Elaborate on the concept of sharding in MongoDB, explaining its purpose and how it works in detail.
 - c. Provide an example-driven explanation of the process of deleting a document in a Collection in MongoDB.
 - d. Write a note summarizing the key aspects of Master/Slave replication in MongoDB, including its significance and configuration.

3. **Attempt any two of the following:** 12
 - a. Identify and discuss the specific fields that are used for sharding in MongoDB.
 - b. Differentiate between the Data Storage Engines used in MongoDB, specifically comparing NMAP and WiredTiger storage engines.
 - c. Evaluate MongoDB's limitations from a security perspective, highlighting potential vulnerabilities and challenges.
 - d. Provide an illustration of "GridFS – The MongoDB File System" using a clear and organized diagram, explaining its components and functioning.

4. **Attempt any two of the following:** 12
 - a. Provide a detailed description of Solid State Disk (SSD) technology, including how it works and its advantages.
 - b. Use a diagram to visually discuss the architecture of Oracle 12c In-Memory Database, highlighting its key components and data flow.
 - c. Explain the concept of an "Event" in jQuery and provide the syntax and usage examples for the fadeIn() and fadeOut() jQuery methods.
 - d. Define jQuery and explain the jQuery element selector, id selector, and class selector, supporting each with relevant examples.

5. Attempt any two of the following:**12**

- a. Identify and describe the purpose and use of the Stringify method in JavaScript.
 - b. Analyze the components and rules that make up the JSON Grammar, providing an in-depth understanding of its structure.
 - c. Describe the process of JSON parsing, including the syntax and how it works in JavaScript.
 - d. Create a visual representation or diagram to explain the structure of the Hypertext Transfer Protocol (HTTP) - Request, detailing its components and their interactions.
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